Topic 3 Review [98 marks]

1. [Maximum mark: 3] 25M.1.SL.TZ2.2

Explain **one** reason why a virtual private network (VPN) might be used. [3]

Markscheme

*Award **max** [3].*
Award [1] for a reason/example/scenario where VPN is relevant.
Award [1] for a feature of VPN (encryption, tunnelling, leak protection, data or bandwidth caps, zero-logs policy, IP shuffling, kill switch). (Accept either a term/name or a description of a feature.)
Award [1] for an elaboration.

Note: *The reason/example/scenario, feature and elaboration must work as whole coherent answer. The following answers are only examples there could be other appropriate answers.*

Employees may use a VPN to access company's network/internal resources/databases/from remote location:
VPN creates a tunnel;
allowing secure connections and access (based on employees' roles);

Users who need a secure/private communication may use VPN;
VPN encrypts data;
and prevents hackers from intercepting sensitive information;

Users of insecure public Wi-Fi/safer browsing while on public Wi-Fi;
logging in on their socials/purchasing something online/leaving credentials;
VPN lets users connect to the provider's servers creating a private tunnel that secures data;

Users may use a VPN to give themselves more privacy/anonymity online/avoid geographical limitations;
IP address is masked so the location of the user is not known;

So they can visit websites that are restricted by location/online businesses use geographic segmentation which means prices of product that can be bought and sold (books, airline tickets, clothes) vary depending on the buyer's location/may be essential if a country is likely to block internet content from foreign entities, for example, journalists could use a VPN to look like they are within that country (*Accept any legitimate reason for needing to be unknown*);

Use of a VPN during days and times when a lot of people trying to stream content and download files at the same time/to avoid congestion (during peak times of internet usage);
With a VPN user can choose what network their traffic travels on;
So, VPNs can (sometimes even) increase the internet speed;

VPNs can be used for bypassing network restrictions;
to access certain blocked websites;
VPNs can tunnel around the restrictions;

Internet/social media users (streaming, online gaming, etc.) may use VPN to avoid tracking/surveillance;
because VPN hides the type of traffic;
and it can prevent the risk of getting frequent advertisements/the risk of bandwidth throttling (slowing down) by ISP provider;

Examiners report

SL:

Only a handful of students achieved full marks on this question. Many referenced a remote working scenario and gained a mark, but instead of focusing on the use of VPNs, they elaborated on remote working itself.

HL:

Most students effectively highlighted key uses of VPNs, illustrating scenarios such as teleworking, secure access, and bypassing geographical or network restrictions.

2. [Maximum mark: 15] 25M.1.SL.TZ2.10

A school has a local area network (LAN) with a central server that stores many files containing personal, health, and financial information.

The LAN is used by the following types of user: network administrators, teachers, students and guests.

(a) Explain how the different levels of access for the users of this LAN could be implemented. [6]

Markscheme

*Award **max** [6].*

Example 1:

Each user should be assigned a user role (administrator, student, teacher, guest);
Each user must have a unique username and password;
Different access levels could be implemented by using password/authentication (PIN/token/biometrics/two-factor authentication (2FA) before accessing a file/through whitelisting/blacklisting devices/by verifying MAC addresses;
Users assigned an administrator role have full access to the system/are allowed to perform most operations on the system (such as, add, remove, edit users, groups, domains, manage passwords, configure services, etc.);
Users assigned a teacher role have read-write access to update student grades/read-only access to students' records/no access to other teachers' financial records;

A user assigned a student role has read access to their reports/no access to other students'/teachers' data;
A user assigned a guest role has restricted/limited access to certain file/folders/information/no access to the files with sensitive or personal data/users with guest accounts can only read from files containing general information/guest users cannot perform any actions/guest users can only browse the internet);

Example 2:

Create a Virtual Local Area Network (VLAN) that allows segmentation of different user groups;
Each VLAN uses strong authentication/user logins/two-factor authentication;
Each VLAN can have specific security policies/rules and restrictions;
Implement Access Control Lists (ACLs) on devices/define what resources each group of users can access/assign permissions based on user roles (RBAC);
Administrators have full access to all resources;
Teachers have access to their personal records/teaching materials/authorized student data;
Students have access only to their own files/assignments/reports/no access to others' data;
Guests have internet access only/restricted access through firewall and ACLs/no access to sensitive information;

Example 3:

Intranet and extranet could be implemented;
The intranet is restricted to group of users related to the school (administrator/teachers/students);
Each teacher/student has login, password (and permissions);
Administrators have full access to all resources (intranet and extranet);
Teachers have access to their personal records/teaching materials/authorized student data;
Students have access only to their own files/assignments/reports;
Extranet is accessible to all (administrator/teacher/student/guest)/Extranet

provides limited access from outside the LAN for all users working remotely;
Guests have access to Internet;

Examiners report

SL:

Several varied responses were seen here ranging from the use of VLAN to extranets. It was a six marker and students were expected to respond with more specificity rather than just mention 'teachers will have more access than students'.

HL:

Majority of the responses included a range of implementation methods, with VLANs, intranet/extranet access, and role-based authentication being commonly mentioned. However, some students did not fully elaborate on user roles and the corresponding privileges. Students are reminded to be mindful of the mark allocation and to develop their points thoroughly to meet the required depth of explanation.

(b.i) Suggest **one** communications link that would provide high-speed internet access for the school. [2]

Markscheme

*Award **max** [2].*
Award [1] for a communication link and [1] for a reason/justification.

Fibre-optic cable;
Because data transmission is fast/reliable;

Coaxial cable (Cable internet);
Because it is (still) one of the fastest internet types/is significantly faster

than DSL/practical because it uses the same coaxial connections as phone services/cable TV;

5G wireless broadband/mobile internet;
Because of greater speed in the transmissions/a greater number of connected devices/the possibility of implementing virtual networks;

Satellite link;
higher data transmission rate/better coverage;

Examiners report

SL:

Very few students provided the correct answer. The most common incorrect response was "ethernet cable." While some mentioned the use of 5G or fibre optics, they failed to justify their choices. Others simply repeated phrases from the question, stating that it provides 'high-speed internet'.

HL:

Many responses correctly identified the appropriate communication link. In some responses which suggested fibre optic cable, the reasoning was often vague — students simply stated that it is 'fast' without specifying that it enables high-speed data transmission. Greater clarity and precision in explanations are needed to secure full marks.

(b.ii) Suggest **two** measures to protect the school's LAN from external network security threats. [4]

Markscheme

*Award **max** [4].*
*Award [1] for a security measure and [1] for justification, **x2**.*

Enable firewalls;
To monitor/prevent unauthorized access to the network/to alert network administrators to any intrusion attempts/to monitor all the incoming and outgoing traffic of the school network/to set the rules to blacklist certain websites;

Router/access point can be configured;
to allow only pre-approved MAC addresses (approved by network administrator) to connect to the network/whitelist/blacklist filtering;

Fibre optics;
To reduce the chances of interception/more challenging for hackers to intercept;

Install new updates/enable automatic updates (OS/firmware/apps, whenever/wherever possible);
to ensure running the latest/most secure versions;

Use encryption;
if cybercriminals gain access to the network, encryption prevents them from reading/understanding any of sensitive information;

Install an antivirus program on all computers;
run or schedule regular virus scans to keep computers virus-free;

Install anti-spyware software;
that records/prevents/removes every attempt/keystroke to gain access to passwords/other sensitive information;

Train/educate users (teachers, students);
to use complex passwords/to update passwords on regular basis/to ignore email messages from unknown parties and never click on links or open attachments;

Use multi-factor authentication (MFA)/two or more different types of actions to verify identity;
to defend against the threat of password-related cyberattacks;

Use of VPN;
ensures secure access through encryption/tunnelling;

A captive portal/implement an authorisation online portal;
Newly connected users should enter password/user ID before they are granted access to network resources;

Note: *Do not accept backup as a method as it does not prevent the threat.*

Note: *Reward other reasonable responses.*

Examiners report

SL:

Several varied answers were seen here with many students achieving a full score.

HL:

A good number of students provided correct response to this question. However, answers that suggested physical security measures or data backups did not earn marks, as these were not relevant to the context of the question.

The school gives students and teachers their own school email account.

(c) List **three** problems that might result from providing email access to all students and teachers. [3]

Markscheme

*Award **max** [3].*

emails may fill the storage available;
emails may overload the internet connection/emails with larger file transfer can utilize a lot of network bandwidth;
phishing attacks/potential data leak – students click link in emails/visit an infected website;
downloading of viruses or worms from emails corrupting the whole network;
if email accounts are compromised, attackers could gain access to school systems/distribute malware across the network;
inappropriate messages may be sent within/outside the school/students may use it for bullying;
e-mail messages could be used to cheat on exams/assignments;
teachers/students may become vulnerable to email from strangers/a risk of sending/receiving offensive/inappropriate/harmful content through email;
inappropriate use of school email for personal communication;
many email accounts can increase the workload for IT staff/more IT support/monitoring/maintenance needed;
increased costs (for licensing, email server, etc.)

Note: *Reward other reasonable answers.*

Examiners report

SL:

This was a question where students were expected to apply their knowledge of privacy and security to a specific example such as an email system in a school. Some students wrote generic one-word answers such as “traffic” or “maintenance” which did not get a mark.

HL:

The responses generally lacked a range of distinct points, with repetition limiting students' ability to earn full marks. Additionally, the question specifically asked for problems from the school's perspective, but this detail was often overlooked. Many students focused instead on issues faced by teachers and students, which did not fully address the requirements of the question.

3. [Maximum mark: 2] 24N.1.SL.TZ0.5

Outline what is meant by a media access control (MAC) address. [2]

Markscheme

Award ***[2 max]***

a physical address/ hardware identification number/ 12-digit hexadecimal number assigned by the manufacturer to a network interface in a device;
that helps to uniquely identify each device on a network;

Examiners report

Most students earned only one mark here. Many described the use of a MAC address rather than explaining what it is.

4. [Maximum mark: 2] 24M.1.SL.TZ1.12

Identify **two** of the layers of the Open Systems Interconnection (OSI) seven-layer model. [2]

Markscheme

Award ***[2 max]***

Physical Layer;
Data Link Layer;
Network (layer);
Transport Layer;
Session Layer;
Presentation Layer;
Application Layer;

Examiners report

This question had a good number of correct responses. Students could write the names of the layers exactly. Very few students did not use the correct term — they wrote "transportation" instead of "transport".

5. [Maximum mark: 15] 24M.1.SL.TZ1.13

Different transmission media may be used within a network.

(a.i) Identify **two** characteristics of fibre optic cables as a transmission medium. [2]

Markscheme

*Award **[2 max]***

Allows very fast transmission of data / Extremely high bandwidth possible;
(Made of glass / plastic fibre) that can transfer information via pulses of light;
Immune to electromagnetic interference / temperature changes / severe weather / highly resistant to noise and moisture;
Very high security rating;

Safe to use in high-voltage locations, areas where flammable gases / chemicals;
Very long distance of transmission before requiring repeaters / attenuation;
They have a long lifespan (thinner and light weighted, so more flexible than other media);
Expensive / harder to install;

Examiners report

Students who were able to give a brief answer stating two characteristics of fibre optic cables achieved maximum marks. Some vague responses (cables are faster / secure) or one-word responses did not score full. The transmission part is missed in these responses.

(a.ii) Identify **two** characteristics of wireless transmission. [2]

Markscheme

Award ***[2 max]***

Uses radio waves / electromagnetic waves to transfer data;
Transmission speed of data is limited;
Range of transmission / transmission reliability can be affected by distance from access point / number of other users/ obstacles etc;
Inexpensive to install/ no need to spend on cabling;
Relatively easy to expand/ add new devices / scale down to accommodate changes in demand;
Security can be poor (unless encryption is applied) / subject to eavesdropping / interception;
Allows users to move around without losing access to the network;

Examiners report

Students who were able to give a brief answer stating two characteristics of wireless transmission achieved maximum marks. Some vague responses (just outlining that the wireless transmission is without wires), or one-word responses did not score full.

(b) Describe how encryption is used to protect data during transmission. [3]

Markscheme

*Award **[3 max]***

Plain text is changed to cypher text / Data is scrambled using an encryption algorithm / key / A Key is required by sender and receiver for authentication;
Cypher text / data cannot be understood if intercepted;
The cypher text/data is then decrypted using a (decryption) key when received by the receiver;

Examiners report

Though the students had an idea of encryption, they did not score allotted maximum marks as they did not use the appropriate keywords. Responses did not include "Key" for encryption and decryption. Students wrote "data cannot be assessed/read by hackers" where they should have written "data cannot be understood by hackers even though they could assess/read".

(c) Explain how data is transmitted using packet switching. [5]

Markscheme

*Award **[5 max]***

The whole data is split into (fixed / equal) sized chunks / packets;
Each packet has a header, payload and trailer;
Packet contains information such as source, destination IP addresses, packet number, protocol, checksum, payload / data, CRC etc (at least two);
Each packet is sent individually / independently along the best path (by a router);
Packets may take different routes to the destination;
If a route becomes unavailable, individual packets can be re-routed;
Packets (may) arrive at the destination out of order;
Packets are re-ordered / joined together at the destination;
Missing packets can be re-sent;

Examiners report

Students who elaborated all the steps in packet switching, right from data split into packets till packets assembled at the destination, achieved the maximum score. It is recommended that the students use the complete technical terms "Destination IP address", "Source IP address" instead of just writing "address".

(d) Explain **one** social implication of changes to working patterns due to the use of a virtual private network (VPN). [3]

Markscheme

*Award **[3 max]***
*Award **[1]** mark for implication (positive or negative) and award **[2]** for further expansion*

Example 1
Improved work / life balance;
Working from home / access to work materials has become much easier /

more secure which can allow more time to be spent at home;
Time spent on travelling to work can now be spent with the family / on hobbies and interests / travelling;
Also, working hours can be flexible, so the employee is less likely to feel stressed leading to improved mental health;

Example 2
It can have a negative impact on work / life balance;
With working from home some workers may feel that their employer has higher expectations of the amount of work they should be doing because of the time they have saved not having to travel to work;
Also, some workers may feel that their employer doesn't trust that they are fully committed to working when at home / distracted by family, so feel under pressure to spend more time on the job/cannot concentrate enough on job, leading to poorer mental health;

Example 3:
Working from home may also result in less collaboration / team work.
As working at different times / zones may not allow employees to communicate effectively with each other;
and feeling of isolation can affect mental health and productivity;

Note: No marks if there is no social implication mentioned.

Examiners report

Many students had difficulty in correctly expressing the social implications. It was observed that the responses included economical, technical implications and some ethical too instead of social implications.

6. [Maximum mark: 3] 23N.1.SL.TZ0.4

State **three** pieces of information that a data packet must contain. [3]

Markscheme

*Award **[3 max]**.*
Data;
Protocol;
Packet number;
Total number of packets;
Sender's IP address;
Receiver's IP address;
Control bits/Parity bit/Check digit/Time to live (hop limit);

Examiners report

SL:

Students who mentioned "Sender/Receiver IP Address" rather than simply "Sender/Receiver address", achieved higher marks.

HL:

Many students identified three pieces of information that a data packet must contain and scored full or nearly full marks for this question.

7. [Maximum mark: 15] 23N.1.SL.TZ0.10

The staff at a doctor's practice consist of a receptionist and a doctor.

The patients' medical records and payments, the doctor's appointment calendar, and other important data are stored in a database on the central computer.

(a) Outline **one** security measure that can be taken to prevent unauthorized access to the patients' data stored on the central computer. [2]

Markscheme

Award ***[2 max]***.

Passwords should be given to access certain aspects of the data;
There should be levels of hierarchy (for example, the receptionist will be only allowed to access data such as names, addresses but not medical history/doctor’s notes);

Multi-factor authentication/one-factor authentication;
which mandates for users (the doctor and receptionist) to verify their identities through various methods of validation (for example, PIN, incorporate thumb scanning or retina scanning technology)/which prevents random persons logging into the system;

Most security breaches are a consequence of human factors (negligence or simple “human” error);
This can be prevented by security awareness training for the doctor and receptionist/precautionary measures when handling patient data;

Encryption;
makes it harder for hackers to decipher confidential patient data (if they manage to breach and subsequently gain access to the information);

Regularly installing updates/patches;
To ensure the data is protected against new threats;

Examiners report

Most students scored at least one mark for mentioning the use of "username and password" but did not provide a suitable expansion. Students who mentioned two different security measures gained only one mark.

(b.i) Identify **one** cause of data loss. [1]

Markscheme

*Award **[1 max]**.*
Human failure/error;
Software corruption;
Theft;
Computer viruses/other malware;
Hardware destruction;
Data corruption;

Examiners report

'Natural disaster' and 'data migration' were the most common incorrect answers; however, students who mentioned 'hardware failure due to a natural disaster' scored a mark.

(b.ii) Describe **one** method that can be used to prevent data loss. [2]

Markscheme

*Award **[2 max]**.*
Offsite/online backups/hard copy backups of patients' records should be made;
To be used in case of system failure;

With data loss prevention software/backup software;
That creates a file back/a copy doctor's files to a secondary device in an automated fashion;

Firewall and antivirus software/all the software;
Need to be updated and properly maintained (to avoid corruption);

Training the staff with security practices;
to avoid human errors;

Physical security;
to protect against theft;

Maintaining the suitable conditions/following best practices for usage;
to avoid hardware destruction;

Note: Reward other reasonable answers.

Examiners report

Many correct answers were given here; backups being the most common one. Students who went on to give a further explanation scored a second mark.

A new vaccine has been distributed that would be of benefit to some of the doctor's patients. A large number of personalized letters need to be written to these patients, inviting them to visit the doctor's practice to be vaccinated.

(c) Describe how these letters could be automatically generated by a word processing application. [3]

Markscheme

*Award **[3 max]**.*
A mail-merge feature can be used to contact them;
The receptionist types the body of the letter (template)/prepares the letter for the patients;
The database containing patient's records can be searched for the names and addresses of this type of patients;
To prepare the mailing list/table of recipients;

Examiners report

Many students were correctly able to mention the use of a template to create a letter and gained one mark but were unable to expand any further. Very few of them were familiar with the process of "Mail Merge".

When the doctor visits a patient in their home, she needs to be able to access the patient's medical records stored on the central computer in the practice.

(d) Outline **two** reasons for the use of a virtual private network (VPN) in this situation. [4]

Markscheme

*Award **[4 max]**.*
Security when working remotely;
VPN's data encryption features (putting data into a coded format so its meaning is obscured) allows the doctor to keep confidential information safe/VPN authenticates the user (doctor) before giving the access to data;

VPN allows access to any content in any place;
Because it enables users to send and receive data across shared or public networks as if their computing devices were directly connected to the private network;

VPNs hide the location/hidden IP address;
making it seem as if she is accessing data from another place/location of the patient unknown for hackers/the MitM;

Protects the privacy of data;
A VPN will prevent apps and websites from attributing the doctor's behaviour to her computer's IP address/can limit the collection of the location and browser history from the Internet Service Provider(ISP);

VPNs usually have intuitive and user-friendly interface;
So, it makes installation and use easy for the doctor (non-technical user);

VPN is adaptable to many smart devices/protects smart devices such as phones, tablets and laptops;
So, the doctor can use various devices/any available device;

*Mark as **[2]** and **[2]**.*

Examiners report

Students who identified security or remote access as uses of VPN and went on to explain it further achieved higher marks. Students who mentioned 'Encryption' should be aware that encrypted text can still be accessed or read, but that it cannot be understood.

A mobile data connection enables the doctor to access internet resources while visiting patients in their homes. Sometimes the data transfer speeds are slow.

(e) Explain why the speed of data transmission across a mobile network can vary. [3]

Markscheme

*Award **[3 max]**.*
The network technologies available in the area (3G, 4G, 5G);
Because of limited bandwidth;
because the features of the doctor's device;
because of the amount of data being transmitted;

because the number of other users in the area using the same network (during certain hours, there are many users, which causes the connection speed to slow down);
because of the location of user with respect to cell towers (the speed may change because the signal varies depending on the coverage area);

Examiners report

Many correct answers were seen to identify factors that may affect speed of data transmission, including amount of traffic, size of data, bandwidth etc. A few students did not realise that the question focused on 'data transmission across a mobile network' and mentioned 'distance from the router' as a factor.

8. [Maximum mark: 15] 23M.1.SL.TZ2.11

Many organizations use a virtual private network (VPN) to enable employees working remotely to access files that are held on the organization's server.

(a) State **two** technologies that are required to provide a virtual private network (VPN). [2]

Markscheme

Award ***[2 max]***
(VPN) tunnelling (server);
(VPN aware) router (and firewall);
Encryption (protocol) *(Accept examples IPSec/ SSL / TSL)*;
VPN client software (installed on the employee's computer);

Examiners report

SL:

Virtually all candidates correctly stated one or two technologies required to provide a VPN from a VON tunnelling server, a VON aware router, an encryption protocol or VPN client software.

HL:

Many responses to (a) and (b) were verbose while missing crucial information that would award the mark.

A few candidates confused a VPN with Wi-Fi.

(b) Identify **two** factors that may affect the speed of data transmission. [2]

Markscheme

*Award **[2 max]***
physical distance/ the number of network devices which have to be crossed;
the performance of each of the devices (between sender and receiver), for example weak processor;
quality/characteristics of network equipment (such as the router or transmission media/cable/fibre/wireless);
number of network users (and their demand at any particular time);

Accept other reasonable answers, such as the type of encryption used or the encryption strength/ server bandwidth/ size of the user data/ type of protocol used, etc.;

Examiners report

SL:

Many correct answers were seen to identify factors that may affect speed of data transmission, including amount of traffic, size of data, physical distance to travel or characteristics of network equipment.

HL:

Many responses to (a) and (b) were verbose while missing crucial information that would award the mark.

(c) Explain why data compression would be used when data is transmitted. [3]

Markscheme

*Award **[3 max]***
(Because) compression reduces the size of a file/ size of data/ the number of packets to be transmitted;
Which reduces transmission time/ consumes less bandwidth;
And can result in significant cost savings;

Examiners report

SL:

Candidates mostly achieved the mark explaining that data compression reduces the size of the data to be transmitted. Some candidates achieved additional marks for noting that the reduced file size would reduce the transmission time or the amount of bandwidth consumed, which can significantly reduce costs.

HL:

A well answered question.

A large amount of sensitive data is stored online and needs to be protected.

(d) Outline how encryption is used to protect data. [2]

Markscheme

*Award **[2 max]***
Encryption scrambles readable text;
So, it can only be read/understood by the person who has the decryption key;

Data encryption translates plain text into ciphertext;
That can be viewed/read in its original form only if it is decrypted with the correct key;

Examiners report

SL:

Candidates who stated that encryption scrambles readable text so that it can only be understood if it is decrypted using the correct key achieved both marks. Candidates should be aware that encrypted text can still be accessed or read, but that it cannot be understood.

HL:

Problems that may occur during data migration were well explained.

(e) Describe the role of a firewall. [2]

Markscheme

*Award **[2 max]***
A firewall monitors incoming and outgoing network traffic;
and decides whether to allow or block specific traffic (based on a defined set of security rules)/ restricts access to parts of a network / prevents unauthorised access of confidential data);

***Note:** Accept software and hardware firewalls.*

A (software) firewall (is a program installed on each computer that) monitors incoming and outgoing traffic/ controls the behaviour of applications;
and filter/block traffic coming from unsecured or suspicious apps/ blacklisted apps;

A (hardware) firewall (is a piece of equipment installed between the network and gateway that) regulates traffic through ports;
and prohibits suspicious data packets from passing through;

Examiners report

SL:

Candidates who described a firewall as monitoring incoming and outgoing network traffic and blocking traffic based on a pre-defined set of rules achieved both marks. Some candidates alternatively correctly identified the fact that a firewall can be used to prevent unauthorised access to a network.

HL:

A well answered question. Candidates seemed to be very familiar with the direct changeover and parallel running.

Employees are increasingly working from home.

(f) Discuss the social impacts of this changed work pattern on employees. [4]

Markscheme

*Award **[4 max]***
*Award **[2 max]** for positive aspects, 1 mark for stating any of the positive aspects (of working from home) and 1 mark for discussion.*
*Award **[2 max]** for negative aspects, 1 mark for stating any of the negative aspects (of working from home) and 1 mark for discussion.*

Examples (Positive aspects):
Flexibility and agility
Increased productivity - due to fewer interruptions
Increased motivation

Autonomy
Improved health and wellbeing
Better work/life balance

Examples (negative aspects):
Working from home doesn't suit everyone
Employees feeling isolated
Home distractions
Negative impact on mental health
Not all jobs suit home working

Note: *When a change to the work pattern is identified it can be an advantage or a disadvantage. For example, working at home can improve interaction with the family but also provide a distraction to work.*

Examiners report

SL:

The vast majority of candidates achieved marks for discussing the social impacts of employees working from home. Candidates who discussed both positive and negative aspects achieved full marks.

HL:

All candidates attempted it and many got it right.

9. [Maximum mark: 4] 22N.1.SL.TZ0.5

A student posts images and videos on a public website of her friends at a party.

(a) Outline **one** ethical issue with the student posting these images and videos. [2]

Markscheme

*Award **[2]** max.*
the video/images could be viewed by anyone;
which may not be desirable since some people may not wish to have themselves regarded by everyone;

anybody could download the images/video;
and they can be manipulated by outsiders;

posting images of other people at a private function requires their permission;
did she get permission from her friends? ;
that they were fine with her posting and share their pictures with everyone;

Examiners report

Most candidates recognised that the ethical issue in the question was related to privacy. Candidates who identified this fact along with an expansion regarding private images being available to everyone achieved both marks.

(b) Outline **one** technical issue that may prevent the images and videos from being viewed. [2]

Markscheme

*Award **[2]** max.*
there could be problems if the person at the event does not have the facility to see the video/images;
because of, for example, inadequate internet connection / browser / video player;

space on web server could cause problems;
if the size of files exceeds the allocation on server;

Examiners report

Most candidates recognised issues related to poor internet connection or equipment being faulty, inappropriate or not working as technical issues that could prevent the images from being viewed. However, some candidates gave answers related to site moderation as preventing the images from being viewed. Whereas this may be true, it isn't a technical issue.

10. [Maximum mark: 15] 22N.1.SL.TZ0.9

A company has a local area network (LAN). Ethernet (a wired network) and WiFi (a wireless network) are the two ways to enable LAN connections.

The LAN is accessible to all employees through their personal accounts. At the office, employees can use either desktop computers for wired access to the LAN or personal laptops to connect wirelessly.

(a) Identify **one** additional hardware component in a wireless LAN. [1]

Markscheme

*Award **[1]** max.*
(Wireless) router;
Access points;
(Wireless) bridge or repeater;
(Wireless) controller;
Wireless adaptors/ Network interface cards (NICs) (that operate within the employees' computers);

Examiners report

Virtually all candidates correctly identified a piece of additional hardware used in a wireless LAN, with the most common response being a router.

(b) Distinguish between a wired network and a wireless network in terms of reliability of transmission. [4]

Markscheme

*Award **[4]** max.*

The reliability of wireless depends on the strength of the wireless signal;
Depends on distance from router;
depends on the topology/shape of the surroundings;
a wireless LAN has slower data transfer;

whilst
Ethernet is more reliable as the strength of the signal does not depend on the distance from the router/ wired LAN support longer distances;
wired is immune to interference;
there is no issue with the topology/shape of the surrounding;
but the Ethernet cable may be cut/broken affecting reliability;

Examiners report

Candidates generally gave good descriptive responses to distinguish between the reliability of wired and wireless networks, specifically in terms of transmission speed and how this may be disrupted by load, distance from the router, etc. Unfortunately, some responses overlapped with question part c, which didn't necessarily match with reliability issues. These candidates didn't always go on to give a full answer to part c, believing a reference to their answer in part b to be sufficient.

(c) Outline why a wireless network may be less secure than a wired network. [2]

Markscheme

*Award **[2]** max.*
wired network is only accessible with a physical cable connection and in wireless networks signals are broadcasted outside of the building;
leaving it open to the public and potential hackers/ it is easy to intercept transmissions;

a wireless network is more open to misuse;
as network administrator cannot directly monitor a specific machine;

Examiners report

Most candidates recognised that wireless networks are more likely to be accessed by intruders due to the nature of the signal being transmitted wider than the company itself.

Employees who are not in the office can access the company's resources over the internet using a virtual private network (VPN).

(d) Outline **two** features of a VPN that make it secure. [4]

Markscheme

*Award **[4]** max.*
Authentication;
Nobody outside the VPN should be able to use the VPN and affect the security of the VPN, so only user name and password authentication is not enough;
Strong user VPN authentication uses various methods, including certificates, one-time passwords and software tokens;

Encryption;
If data intercepted, it will not be readable;

encrypting each encapsulated data packet's content with an encryption key (the key is shared only between the VPN's server and clients);

Tunnelling software;
A VPN hides a user's data by encrypting it with a tunnel created between the user's device and the VPN's web server (The user then takes on the IP address of the web server (rather than their true IP), and this leads to one advantage of a VPN (namely that a user can appear to be in a different geographic location than they are actually located in);

Multiple exit nodes;
Makes it hard to distinguish where the data sent is originated (protecting privacy);

*Mark as **[2]** and **[2]***
*Award **[1]** for a feature and **[1]** for a reasonable expansion, x2.*

Examiners report

Candidates clearly understand the benefits and features of a VPN. However, this question required candidates to outline two features, which meant that they should state the features, e.g., authentication, encryption, tunnelling or multiple exit nodes, and then, in each case, provide an expansion. Candidates whose responses could be interpreted in that manner achieved the highest marks.

(e) Packet switching is used for transmitting data.

Explain how data is transmitted by packet switching. [4]

Markscheme

*Award **[4]** max.*
Data is organized in specially formatted units (data packets);
Each data packet contains: data, address of the sender, address of the

receiver, size of packet, sequence number, control codes, etc.;
which are routed from source to destination using network switches and routers;
network switches and routers determine how best to transfer the packet between a number of intermediate devices (routers and switches) on the path to its destination (rather than flowing directly over a single wire on the path to its destination);
data packets are reassembled at destination;

Examiners report

Candidates often, but sometimes with a few inaccuracies, gave good descriptions of the process of packet switching.

11. [Maximum mark: 2] 22M.1.SL.TZ0.3

Identify **two** differences between a wide area network (WAN) and a local area network (LAN). [2]

Markscheme

*Award **[2 max]**.*
WAN covers a much larger area (national/international), LANs usually cover a smaller area (such as a single site);
Nodes connected to WANs often make use of connections through public networks (such as the telephone system), nodes connected to LANs are usually connected through private infrastructure;
LANs are more secure than WANs (due to how WANs transmit the data /how far the data would need to travel);
A higher bandwidth is available for transmission in a LAN than a WAN/ LANs can have a higher data transfer rate than a WAN;
LAN is typically cheaper than WAN to implement/ maintain (as the equipment required for LAN is less expensive);
WAN includes a large number of devices connected together, LAN includes

less;
WANs are typically slower than LANs due to the distance data must travel;
WAN requires hardware to connect different networks, such as a router, LAN can be simple and does not need to connect to other networks;

Examiners report

Candidates were aware of the differences between wide area networks and local area networks, but often spent a long time writing a description that only covered one difference, rather than the two required by the question.

12. [Maximum mark: 2] 22M.1.SL.TZ0.4

Outline the reason for compression when transmitting data. [2]

Markscheme

*Award **[2 max]**.*

The reason for compression when transmitting data is to save on transfer times;
As it reduces the number of bits needed to represent data (when compared with the original data);

Compressing data involves modifying/restructuring files, so that they take up less space;
And this results in cost savings in cloud storage;

To take up less bandwidth;
Because data compression reduces the size of files to be transmitted over a network;

*Note to examiners: Award **[1]** for a reason (for example, to save data usage for sending files over the internet, to save storage capacity, to speed up file transfer, to decrease costs for network bandwidth, etc.), and award **[1]** for an expansion.*

Examiners report

Candidates were aware of a reason for the use of compression when transmitting data. Some candidates, however, directed their responses to generic reasons for compression, rather than the scenario of data transmission, as stated in the question.

13. [Maximum mark: 2] 22M.1.SL.TZ0.10

Outline **one** reason for the use of standards in the construction of networks. [2]

Markscheme

Award ***[2 max]***

Standards ensure compatibility between nodes on the network;

Through the use of common techniques/protocols/language;

Examiners report

Most candidates recognised that the use of standards in the construction of networks was related to the need for compatibility, with many answers based on that concept seen.

14. [Maximum mark: 2] 21M.1.SL.TZ0.6

Identify **two** reasons why fibre optic cable would be preferred over wireless connectivity. [2]

Markscheme

Award ***[2 max]***
Fibre optics allow faster transmission speeds;
Fibre optic cables are more secure/harder to break into;
Fibre optic cable transmission is more reliable/less likely to suffer interference;
Fibre optics allow transmission over longer distance;
Fibre optics allow greater bandwidth;

Examiners report

This question was answered very well with the vast majority of candidates able to give two reasons why fibre optic cables would be preferred over a wireless connection.

15. [Maximum mark: 1] 21M.1.SL.TZ0.9

Define the term *data packet*. [1]

Markscheme

Award ***[1 max]***
A unit of data made into a single unit (that travels along a given network path);
A data packet travels through a network as a unit i.e. with all parts kept together;

Examiners report

Candidates generally understood what is meant by data packet and gave a sufficient response to achieve the mark.